

Definition .0.1

Let K be a field. An *absolute value* on K is a function

$$|\cdot| : K \rightarrow \mathbb{R}$$

such that for all $x, y \in K$:

- (1) $|x| \geq 0$, and $|x| = 0$ if and only if $x = 0$;
- (2) $|xy| = |x||y|$;
- (3) $|x + y| \leq |x| + |y|$.

Definition .0.2

An absolute value $|\cdot|$ on K is called *non-archimedean* if condition (3) can be strengthened to

$$|x + y| \leq \max\{|x|, |y|\} \quad \text{for all } x, y \in K.$$

Otherwise, $|\cdot|$ is called *archimedean*.

Remark .0.1

Any absolute value on K defines a metric on K by

$$d(x, y) = |x - y|.$$

Thus every absolute value induces a topology on K .

Definition .0.3

Let $|\cdot|_1$ and $|\cdot|_2$ be absolute values on K . We say they are *equivalent*, and write

$$|\cdot|_1 \sim |\cdot|_2,$$

if they induce the same topology on K .

Example .0.1

The *trivial absolute value* on K is defined by

$$|x|_{\text{triv}} = \begin{cases} 0, & x = 0, \\ 1, & x \neq 0. \end{cases}$$

This is a non-archimedean absolute value.

Definition .0.4

A *place* of K is an equivalence class of nontrivial absolute values on K .

A place is called *infinite* if its absolute values are archimedean, and *finite* otherwise.

Proposition .0.1

Let $|\cdot|$ and $|\cdot|'$ be absolute values on K . Then the following are equivalent:

- (1) $|\cdot| \sim |\cdot|'$;
- (2) for every $x \in K$,

$$|x| < 1 \iff |x|' < 1;$$

(3) there exists $s > 0$ such that for all $x \in K$,

$$|x|' = |x|^s.$$

Proof. (3) $\Rightarrow$ (1). If $|x|' = |x|^s$ for some $s > 0$, then the two absolute values define the same open balls, hence the same topology. So they are equivalent.

(1) $\Rightarrow$ (2). Suppose $|\cdot|$ and $|\cdot|'$ induce the same topology.

We claim that

$$|x| < 1 \iff x^n \rightarrow 0 \text{ with respect to } |\cdot|.$$

Indeed, if $|x| < 1$, then $|x^n| = |x|^n \rightarrow 0$. Conversely, if $x^n \rightarrow 0$, then necessarily $|x| < 1$, since if $|x| \geq 1$, then $|x^n| = |x|^n$ does not tend to 0.

Because the two topologies are the same, a sequence converges to 0 with respect to $|\cdot|$ if and only if it converges to 0 with respect to $|\cdot|'$. Therefore

$$|x| < 1 \iff x^n \rightarrow 0 \text{ in } |\cdot| \iff x^n \rightarrow 0 \text{ in } |\cdot|' \iff |x|' < 1.$$

(2) $\Rightarrow$ (3). First note that (2) also implies

$$|x| > 1 \iff |x|' > 1,$$

since $|x| > 1$ is equivalent to $|x^{-1}| < 1$, and similarly for $|\cdot|'$.

Also, $|\cdot|$ is trivial if and only if $|\cdot|'$ is trivial. Indeed, in the trivial case every nonzero element has value 1, and this is detected by the conditions $|x| < 1$ and $|x| > 1$ never occurring for $x \neq 0$.

So we may assume both absolute values are nontrivial. Choose $y \in K$ such that

$$0 < |y| < 1.$$

Fix $x \in K^\times$. Since $|y| \neq 1$, there exists a unique real number α such that

$$|x| = |y|^\alpha.$$

Now let $\frac{m_i}{n_i}$ be a sequence of rational numbers converging to α from above. Then

$$|x| = |y|^\alpha > |y|^{m_i/n_i}.$$

Since $0 < |y| < 1$, raising to larger exponents makes the value smaller, so after rearranging we get

$$|x|^{n_i} > |y|^{m_i}.$$

Hence

$$\left| \frac{x^{n_i}}{y^{m_i}} \right| > 1.$$

By (2), this implies

$$\left| \frac{x^{n_i}}{y^{m_i}} \right|' > 1,$$

so

$$|x|'^{n_i} > |y|'^{m_i}.$$

Therefore

$$|x|' > |y|'^{m_i/n_i}.$$

Letting $i \rightarrow \infty$, we obtain

$$|x|' \geq |y|'^\alpha.$$

Similarly, taking a sequence $\frac{m_i}{n_i} \rightarrow \alpha$ from below, the same argument gives

$$|x|' \leq |y|'^\alpha.$$

Hence

$$|x|' = |y|'^\alpha.$$

Since $|x| = |y|^\alpha$, taking logarithms gives

$$\alpha = \frac{\log |x|}{\log |y|},$$

and thus

$$\log |x|' = \alpha \log |y|' = \frac{\log |y|'}{\log |y|} \log |x|.$$

Set

$$s = \frac{\log |y|'}{\log |y|} > 0.$$

Then

$$\log |x|' = s \log |x|,$$

so

$$|x|' = |x|^s$$

for all $x \in K^\times$. This is also true for $x = 0$, so (3) follows. $\mathbb{Z}$

Lemma .0.2

An absolute value $|\cdot|$ on K is non-archimedean if and only if there exists a constant $N > 0$ such that

$$|n| \leq N \quad \text{for all } n \in \mathbb{N}.$$

Proof. ($\Rightarrow$) Suppose $|\cdot|$ is non-archimedean. Then

$$|2| = |1 + 1| \leq \max\{|1|, |1|\} = 1.$$

By induction,

$$|n| \leq 1 \quad \text{for all } n \in \mathbb{N}.$$

So we may take $N = 1$.

($\Leftarrow$) Now suppose there exists $N > 0$ such that

$$|n| \leq N \quad \text{for all } n \in \mathbb{N}.$$

We will prove that $|\cdot|$ is non-archimedean.

For any $x, y \in K$ and any integer $n \geq 1$, by the binomial theorem,

$$(x + y)^n = \sum_{i=0}^n \binom{n}{i} x^i y^{n-i}.$$

Taking absolute values and using the triangle inequality,

$$|x + y|^n \leq \sum_{i=0}^n \left| \binom{n}{i} x^i y^{n-i} \right|.$$

Since $\binom{n}{i} \in \mathbb{N}$, we have

$$\left| \binom{n}{i} \right| \leq N.$$

Therefore,

$$|x + y|^n \leq \sum_{i=0}^n N |x|^i |y|^{n-i} \leq (n + 1)N \max\{|x|, |y|\}^n.$$

Taking n th roots,

$$|x + y| \leq ((n + 1)N)^{1/n} \max\{|x|, |y|\}.$$

Now let $n \rightarrow \infty$. Since $((n + 1)N)^{1/n} \rightarrow 1$, we obtain

$$|x + y| \leq \max\{|x|, |y|\}.$$

Thus $|\cdot|$ is non-archimedean. $\mathbb{Z}$

Theorem .0.3 (Product Formula for $\mathbb{Q}$)

For every $x \in \mathbb{Q}^\times$,

$$|x|_\infty \prod_{p \text{ prime}} |x|_p = 1,$$

where $|\cdot|_\infty$ is the usual absolute value on $\mathbb{Q}$ and $|\cdot|_p$ is the p -adic absolute value.